PR 12_1

1. Conexión con el servidor

- Importamos las librerias

In [89]:
```
import mysql.connector
from getpass import getpass
```

- Creamos la conexión

In [90]:
```
try:
    connection = mysql.connector.connect(
        host="mysql-python.cd0w2auewbas.us-east-1.rds.amazonaws.com",
        port=3306,
        user="admin",
        password="12345678"
    )
    print("Conexión exitosa")
except mysql.connector.Error as e:
    print(e)
```

```
Conexión exitosa
```

2. Crear base de datos

- Creamos la base de datos

In [91]:
```
create_empleados_database_query = """
    CREATE DATABASE IF NOT EXISTS empleados;
"""
```

- La utilizamos

In [92]:
```
use_empleados_query = """
    USE empleados;
"""
```

- Creamos la tabla dept

In [93]:
```
create_dept_table_query = """
    CREATE TABLE dept(
        deptno INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
```

```
        dname varchar(255),
        loc varchar(255)
    );
"""
```

- Creamos la tabla salgrade

In [94]:
```
create_salgrade_table_query = """
    CREATE TABLE salgrade(
        grade INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
        losal INT,
        hisal INT
    );
"""
```

- Creamos la tabla emp

In [95]:
```
create_emp_table_query = """
    CREATE TABLE emp(
        eno INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
        ename varchar(255),
        job varchar(255),
        mgr INT,
        hiredate DATE,
        sal INT,
        comm INT,
        deptno INT,

        FOREIGN KEY (mgr) REFERENCES emp(eno),
        FOREIGN KEY (sal) REFERENCES salgrade(grade),
        FOREIGN KEY (deptno) REFERENCES dept(deptno)

    );
"""
```

- Ejecutamos todas las query

In [96]:
```
with connection.cursor() as cursor:
    cursor.execute("DROP DATABASE IF EXISTS empleados;")
    cursor.execute(create_empleados_database_query)
    cursor.execute(use_empleados_query)
    cursor.execute(create_dept_table_query)
    cursor.execute(create_salgrade_table_query)
    cursor.execute(create_emp_table_query)
    connection.commit()
```

3. Insertamos los datos

- Datos de dept

In [97]:
```
dept_data = [
    (10, 'ACCOUNTING', 'NEW YORK'),
    (20, 'RESEARCH', 'DALLAS'),
    (30, 'SALES', 'CHICAGO'),
    (40, 'OPERATIONS', 'BOSTON')
]
```

- Datos de salgrade

In [98]:
```
salgrade_data = [
    (1, 700, 1200),
    (2, 1201, 1400),
    (3, 1401, 2000),
    (4, 2001, 3000),
    (5, 3001, 9999)
]
```

- Datos de emp

In [99]:
```
emp_data = [
    (7369, 'SMITH',  'CLERK',     7902, '1980-12-17',  800,  None, 20),
    (7499, 'ALLEN',  'SALESMAN',  7698, '1981-02-20', 1600,   300, 30),
    (7521, 'WARD',   'SALESMAN',  7698, '1981-02-22', 1250,   500, 30),
    (7566, 'JONES',  'MANAGER',   7839, '1981-04-02', 2975,  None, 20),
    (7654, 'MARTIN', 'SALESMAN',  7698, '1981-09-28', 1250,  1400, 30),
    (7698, 'BLAKE',  'MANAGER',   7839, '1981-05-01', 2850,  None, 30),
    (7782, 'CLARK',  'MANAGER',   7839, '1981-06-09', 2450,  None, 10),
    (7788, 'SCOTT',  'ANALYST',   7566, '1982-12-09', 3000,  None, 20),
    (7839, 'KING',   'PRESIDENT', None, '1981-11-17', 5000,  None, 10),
    (7844, 'TURNER', 'SALESMAN',  7698, '1981-09-08', 1500,     0, 30),
    (7876, 'ADAMS',  'CLERK',     7788, '1983-01-12', 1100,  None, 20),
    (7900, 'JAMES',  'CLERK',     7698, '1981-12-03',  950,  None, 30),
    (7902, 'FORD',   'ANALYST',   7566, '1981-12-03', 3000,  None, 20),
    (7934, 'MILLER', 'CLERK',     7782, '1982-01-23', 1300,  None, 10)
]
```

- Introducimos los datos (desactivando y reactivando las restricciones)

In [100…
```
with connection.cursor() as cursor:
    # Desactivamo las restricciones
    cursor.execute("SET FOREIGN_KEY_CHECKS = 0;")

    # DEPT
    cursor.executemany("INSERT INTO dept(deptno, dname, loc) VALUES(%s, %s,

    # SALGRADE
    cursor.executemany("INSERT INTO salgrade(grade, losal, hisal) VALUES(%s,

    # EMP
    cursor.executemany("INSERT INTO emp(eno, ename, job, mgr, hiredate, sal,
```

```
    # Reactivamos las restricciones
    cursor.execute("SET FOREIGN_KEY_CHECKS = 1;")

    # Ingresamos los datos
    connection.commit()
```

4. Consultas

Definimos una función para ejecutar las consultas

In [101…
```
def consultar(consulta):
    with connection.cursor() as cursor:
        cursor.execute(consulta)
        rows = cursor.fetchmany(100)
        for row in rows:
            print(row)
```

1. Seleccionar el nº de empleado, salario, comisión, nº de departamento y fecha de la tabla EMP.

In [102…
```
consultar("SELECT eno, sal, comm, deptno, hiredate FROM emp;")
```

```
(7369, 800, None, 20, datetime.date(1980, 12, 17))
(7499, 1600, 300, 30, datetime.date(1981, 2, 20))
(7521, 1250, 500, 30, datetime.date(1981, 2, 22))
(7566, 2975, None, 20, datetime.date(1981, 4, 2))
(7654, 1250, 1400, 30, datetime.date(1981, 9, 28))
(7698, 2850, None, 30, datetime.date(1981, 5, 1))
(7782, 2450, None, 10, datetime.date(1981, 6, 9))
(7788, 3000, None, 20, datetime.date(1982, 12, 9))
(7839, 5000, None, 10, datetime.date(1981, 11, 17))
(7844, 1500, 0, 30, datetime.date(1981, 9, 8))
(7876, 1100, None, 20, datetime.date(1983, 1, 12))
(7900, 950, None, 30, datetime.date(1981, 12, 3))
(7902, 3000, None, 20, datetime.date(1981, 12, 3))
(7934, 1300, None, 10, datetime.date(1982, 1, 23))
```

2. Seleccionar todas las columnas de la tabla DEPT.

In [103…
```
consultar("SELECT * FROM dept;")
```

```
(10, 'ACCOUNTING', 'NEW YORK')
(20, 'RESEARCH', 'DALLAS')
(30, 'SALES', 'CHICAGO')
(40, 'OPERATIONS', 'BOSTON')
```

3. Seleccionar los nombres y los empleos de todos los empleados, ordenados por empleo.

In [104…
```
consultar("SELECT ename, job FROM emp ORDER BY job;")
```

```
('SCOTT', 'ANALYST')
('FORD', 'ANALYST')
('SMITH', 'CLERK')
('ADAMS', 'CLERK')
('JAMES', 'CLERK')
('MILLER', 'CLERK')
('JONES', 'MANAGER')
('BLAKE', 'MANAGER')
('CLARK', 'MANAGER')
('KING', 'PRESIDENT')
('ALLEN', 'SALESMAN')
('WARD', 'SALESMAN')
('MARTIN', 'SALESMAN')
('TURNER', 'SALESMAN')
```

4. Seleccionar los empleos que hay en cada departamento, ordenados por departamento.

In [105…
```
consultar("SELECT job, deptno FROM emp ORDER BY deptno;")
```

```
('MANAGER', 10)
('PRESIDENT', 10)
('CLERK', 10)
('CLERK', 20)
('MANAGER', 20)
('ANALYST', 20)
('CLERK', 20)
('ANALYST', 20)
('SALESMAN', 30)
('SALESMAN', 30)
('SALESMAN', 30)
('MANAGER', 30)
('SALESMAN', 30)
('CLERK', 30)
```

5. Seleccionar los distintos departamentos que existen en la tabla EMP.

In [106…
```
consultar("SELECT DISTINCT deptno FROM emp;")
```

```
(10,)
(20,)
(30,)
```

6. Calcular el salario anual a percibir por cada empleado.

In [107…
```
consultar("SELECT ename, (sal * 52.1429) AS 'salario_anual' FROM emp ORDER E
```

```
('SMITH', Decimal('41714.3200'))
('ALLEN', Decimal('83428.6400'))
('WARD', Decimal('65178.6250'))
('JONES', Decimal('155125.1275'))
('MARTIN', Decimal('65178.6250'))
('BLAKE', Decimal('148607.2650'))
('CLARK', Decimal('127750.1050'))
('SCOTT', Decimal('156428.7000'))
('KING', Decimal('260714.5000'))
('TURNER', Decimal('78214.3500'))
('ADAMS', Decimal('57357.1900'))
('JAMES', Decimal('49535.7550'))
('FORD', Decimal('156428.7000'))
('MILLER', Decimal('67785.7700'))
```

7. Mostrar el nombre del empleado y una columna que contenga el salario multiplicado por la comisión cuya cabecera sea “BONO".

In [108…

```
consultar("""
    SELECT ename, (sal * 52.1429 * comm) AS 'salario_anual' FROM emp
    WHERE comm IS NOT NULL
    ORDER BY 'salario_anual';
""")
```

```
('ALLEN', Decimal('25028592.0000'))
('WARD', Decimal('32589312.5000'))
('MARTIN', Decimal('91250075.0000'))
('TURNER', Decimal('0.0000'))
```

8. Seleccionar aquellos empleados que sean "SALESMAN".

In [109…

```
consultar("SELECT * FROM emp WHERE job = 'SALESMAN';")
```

```
(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7521, 'WARD', 'SALESMAN', 7698, datetime.date(1981, 2, 22), 1250, 500, 30)
(7654, 'MARTIN', 'SALESMAN', 7698, datetime.date(1981, 9, 28), 1250, 1400, 3
0)
(7844, 'TURNER', 'SALESMAN', 7698, datetime.date(1981, 9, 8), 1500, 0, 30)
```

9. Seleccionar aquellos empleados que no trabajen en el departamento 30.

In [110…

```
consultar("SELECT * FROM emp WHERE deptno != 30;")
```

```
(7782, 'CLARK', 'MANAGER', 7839, datetime.date(1981, 6, 9), 2450, None, 10)
(7839, 'KING', 'PRESIDENT', None, datetime.date(1981, 11, 17), 5000, None, 1
0)
(7934, 'MILLER', 'CLERK', 7782, datetime.date(1982, 1, 23), 1300, None, 10)
(7369, 'SMITH', 'CLERK', 7902, datetime.date(1980, 12, 17), 800, None, 20)
(7566, 'JONES', 'MANAGER', 7839, datetime.date(1981, 4, 2), 2975, None, 20)
(7788, 'SCOTT', 'ANALYST', 7566, datetime.date(1982, 12, 9), 3000, None, 20)
(7876, 'ADAMS', 'CLERK', 7788, datetime.date(1983, 1, 12), 1100, None, 20)
(7902, 'FORD', 'ANALYST', 7566, datetime.date(1981, 12, 3), 3000, None, 20)
```

10. Seleccionar el nombre de aquellos empleados que ganen más de 2000.

In [111…
```
consultar("SELECT ename FROM emp WHERE sal > 2000;")
```

```
('CLARK',)
('BLAKE',)
('JONES',)
('SCOTT',)
('FORD',)
('KING',)
```

11. Seleccionar aquellos empleados que hayan entrado antes del 1/1/82

In [112…
```
consultar("SELECT * FROM emp WHERE hiredate < '1982-01-01';")
```

```
(7369, 'SMITH', 'CLERK', 7902, datetime.date(1980, 12, 17), 800, None, 20)
(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7521, 'WARD', 'SALESMAN', 7698, datetime.date(1981, 2, 22), 1250, 500, 30)
(7566, 'JONES', 'MANAGER', 7839, datetime.date(1981, 4, 2), 2975, None, 20)
(7654, 'MARTIN', 'SALESMAN', 7698, datetime.date(1981, 9, 28), 1250, 1400, 3
0)
(7698, 'BLAKE', 'MANAGER', 7839, datetime.date(1981, 5, 1), 2850, None, 30)
(7782, 'CLARK', 'MANAGER', 7839, datetime.date(1981, 6, 9), 2450, None, 10)
(7839, 'KING', 'PRESIDENT', None, datetime.date(1981, 11, 17), 5000, None, 1
0)
(7844, 'TURNER', 'SALESMAN', 7698, datetime.date(1981, 9, 8), 1500, 0, 30)
(7900, 'JAMES', 'CLERK', 7698, datetime.date(1981, 12, 3), 950, None, 30)
(7902, 'FORD', 'ANALYST', 7566, datetime.date(1981, 12, 3), 3000, None, 20)
```

12. Mostrar el nombre del empleado y su fecha de alta en la empresa de los empleados que son “ANALISTA".

In [113…
```
consultar("SELECT ename, hiredate FROM emp WHERE job = 'ANALYST';")
```

```
('SCOTT', datetime.date(1982, 12, 9))
('FORD', datetime.date(1981, 12, 3))
```

13. Seleccionar los empleados cuyo salario sea superior al de “ADAMS".

In [114…
```
consultar("""
    SELECT * FROM emp WHERE sal > (
        SELECT sal FROM emp WHERE ename = 'ADAMS'
    );
""")
```

```
(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7521, 'WARD', 'SALESMAN', 7698, datetime.date(1981, 2, 22), 1250, 500, 30)
(7566, 'JONES', 'MANAGER', 7839, datetime.date(1981, 4, 2), 2975, None, 20)
(7654, 'MARTIN', 'SALESMAN', 7698, datetime.date(1981, 9, 28), 1250, 1400, 3
0)
(7698, 'BLAKE', 'MANAGER', 7839, datetime.date(1981, 5, 1), 2850, None, 30)
(7782, 'CLARK', 'MANAGER', 7839, datetime.date(1981, 6, 9), 2450, None, 10)
(7788, 'SCOTT', 'ANALYST', 7566, datetime.date(1982, 12, 9), 3000, None, 20)
(7839, 'KING', 'PRESIDENT', None, datetime.date(1981, 11, 17), 5000, None, 1
0)
(7844, 'TURNER', 'SALESMAN', 7698, datetime.date(1981, 9, 8), 1500, 0, 30)
(7902, 'FORD', 'ANALYST', 7566, datetime.date(1981, 12, 3), 3000, None, 20)
(7934, 'MILLER', 'CLERK', 7782, datetime.date(1982, 1, 23), 1300, None, 10)
```

14. Seleccionar los empleados que trabajan en el mismo departamento que "CLARK".

In [115…

```
consultar("""
    SELECT * FROM emp WHERE deptno = (
        SELECT deptno FROM emp WHERE ename = 'CLARK'
    );
""")
```

```
(7782, 'CLARK', 'MANAGER', 7839, datetime.date(1981, 6, 9), 2450, None, 10)
(7839, 'KING', 'PRESIDENT', None, datetime.date(1981, 11, 17), 5000, None, 1
0)
(7934, 'MILLER', 'CLERK', 7782, datetime.date(1982, 1, 23), 1300, None, 10)
```

15. Encontrar a los empleados cuyo jefe es “BLAKE".

In [116…

```
consultar("""
    SELECT * FROM emp WHERE mgr = (
        SELECT eno FROM emp WHERE ename = 'BLAKE'
    );
""")
```

```
(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7521, 'WARD', 'SALESMAN', 7698, datetime.date(1981, 2, 22), 1250, 500, 30)
(7654, 'MARTIN', 'SALESMAN', 7698, datetime.date(1981, 9, 28), 1250, 1400, 3
0)
(7844, 'TURNER', 'SALESMAN', 7698, datetime.date(1981, 9, 8), 1500, 0, 30)
(7900, 'JAMES', 'CLERK', 7698, datetime.date(1981, 12, 3), 950, None, 30)
```

16. Seleccionar el nombre de los vendedores que ganen más de 1500.

In [117…

```
consultar("SELECT ename FROM emp WHERE job = 'SALESMAN' AND sal > 1500;")
```

```
('ALLEN',)
```

17. Seleccionar aquellos empleados que tienen comisión.

In [118…

```
consultar("SELECT * FROM emp WHERE comm IS NOT NULL;")
```

```
(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7521, 'WARD', 'SALESMAN', 7698, datetime.date(1981, 2, 22), 1250, 500, 30)
(7654, 'MARTIN', 'SALESMAN', 7698, datetime.date(1981, 9, 28), 1250, 1400, 3
0)
(7844, 'TURNER', 'SALESMAN', 7698, datetime.date(1981, 9, 8), 1500, 0, 30)
```

18. Seleccionar aquellos que se llamen “SMITH", "ALLEN" o "SCOTT".

In [119…
```
consultar("SELECT * FROM emp WHERE ename = 'SMITH' OR ename = 'ALLEN' OR ena
```

```
(7369, 'SMITH', 'CLERK', 7902, datetime.date(1980, 12, 17), 800, None, 20)
(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7788, 'SCOTT', 'ANALYST', 7566, datetime.date(1982, 12, 9), 3000, None, 20)

(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7788, 'SCOTT', 'ANALYST', 7566, datetime.date(1982, 12, 9), 3000, None, 20)
```

19. Seleccionar aquellos que no se llamen "SMITH", "ALLEN" o "SCOTT".

In [120…
```
consultar("SELECT * FROM emp WHERE NOT(ename = 'SMITH' OR ename = 'ALLEN' OR
```

```
(7521, 'WARD', 'SALESMAN', 7698, datetime.date(1981, 2, 22), 1250, 500, 30)
(7566, 'JONES', 'MANAGER', 7839, datetime.date(1981, 4, 2), 2975, None, 20)
(7654, 'MARTIN', 'SALESMAN', 7698, datetime.date(1981, 9, 28), 1250, 1400, 3
0)
(7698, 'BLAKE', 'MANAGER', 7839, datetime.date(1981, 5, 1), 2850, None, 30)
(7782, 'CLARK', 'MANAGER', 7839, datetime.date(1981, 6, 9), 2450, None, 10)
(7839, 'KING', 'PRESIDENT', None, datetime.date(1981, 11, 17), 5000, None, 1
0)
(7844, 'TURNER', 'SALESMAN', 7698, datetime.date(1981, 9, 8), 1500, 0, 30)
(7876, 'ADAMS', 'CLERK', 7788, datetime.date(1983, 1, 12), 1100, None, 20)
(7900, 'JAMES', 'CLERK', 7698, datetime.date(1981, 12, 3), 950, None, 30)
(7902, 'FORD', 'ANALYST', 7566, datetime.date(1981, 12, 3), 3000, None, 20)
(7934, 'MILLER', 'CLERK', 7782, datetime.date(1982, 1, 23), 1300, None, 10)
```

20. Seleccionar los empleados que trabajen en "CHICAGO".

In [121…
```
consultar("""
    SELECT * FROM emp WHERE deptno = (
        SELECT deptno FROM dept WHERE loc = 'CHICAGO'
    );
""")
```

```
(7499, 'ALLEN', 'SALESMAN', 7698, datetime.date(1981, 2, 20), 1600, 300, 30)
(7521, 'WARD', 'SALESMAN', 7698, datetime.date(1981, 2, 22), 1250, 500, 30)
(7654, 'MARTIN', 'SALESMAN', 7698, datetime.date(1981, 9, 28), 1250, 1400, 3
0)
(7698, 'BLAKE', 'MANAGER', 7839, datetime.date(1981, 5, 1), 2850, None, 30)
(7844, 'TURNER', 'SALESMAN', 7698, datetime.date(1981, 9, 8), 1500, 0, 30)
(7900, 'JAMES', 'CLERK', 7698, datetime.date(1981, 12, 3), 950, None, 30)
```

21. Seleccionar aquellos empleados que trabajen en el departamento 10 o en el 20.

In [122…

```
consultar("SELECT * FROM emp WHERE deptno = 10 or deptno = 20;")
```

```
(7782, 'CLARK', 'MANAGER', 7839, datetime.date(1981, 6, 9), 2450, None, 10)
(7839, 'KING', 'PRESIDENT', None, datetime.date(1981, 11, 17), 5000, None, 1
0)
(7934, 'MILLER', 'CLERK', 7782, datetime.date(1982, 1, 23), 1300, None, 10)
(7369, 'SMITH', 'CLERK', 7902, datetime.date(1980, 12, 17), 800, None, 20)
(7566, 'JONES', 'MANAGER', 7839, datetime.date(1981, 4, 2), 2975, None, 20)
(7788, 'SCOTT', 'ANALYST', 7566, datetime.date(1982, 12, 9), 3000, None, 20)
(7876, 'ADAMS', 'CLERK', 7788, datetime.date(1983, 1, 12), 1100, None, 20)
(7902, 'FORD', 'ANALYST', 7566, datetime.date(1981, 12, 3), 3000, None, 20)
```